FEAT-ZD-48600 · DE-05 Data Model Changes · Data Model change for the feature

Data Engineering agent · item feat.data_model_change · source hobasa-data-engineering-agent/outputs/FEAT-ZD-48600/feat.data_model_change.md · approved by Data engineer sim.data_engineer

feat.data_model_change: FEAT-ZD-48600

Data Model change for the Data Engineering lane of FEAT-ZD-48600 ("Show which overdue invoices belong to vendors that usually pay late"). This is DE-05, Data Model Changes, attempt 2. The workbook action is "Update the data model based on the source systems and ingested data". There is no gate: automated checks, the independent judge and sampled human evaluation review this output (record:brief).

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DE-05 Data Model Changes, attempt 2	record:brief
Inputs	task.components (a126bc4, from DE-04), org.data_model (a6b3e2a)	record:brief
Background read	journey:feat.design (DES-02 output, approved by the Architect and UX lead on the journey board: DC-1, DC-2, B1.2 to B1.4, DD-6, DD-7)	journey:feat.design; record:memory (JOURNEY.md)
Code read	ap_pipeline source and tests, as given in the step's tool view	tool:repo
Source system	Vendor invoice feed (CSV with a header row), v2 today, v3 adds the optional `paid_date` column	task.components (IN-1); org.data_model
Result	**The registered Invoice entity already holds `paid_date` (date or None); this step confirms it field by field against the feed and the code, and adds two items the registered model does not yet hold: the reject reason value `paid_date` on the Rejected row, and the derived Delay history structure produced by `delays_by_vendor`.** No field is removed or retyped; no storage or schema migration is involved	org.data_model; task.components (section 5, TC-1, TC-2, IN-6); journey:feat.design (A1, B1.3)

1. Baseline: registered model and what the code holds today

Element	Registered model (org.data_model)	Code today (`transform.to_model`)	Gap	Source
Invoice `id`	str	`r["invoice_id"].strip()`	None	org.data_model; tool:repo (`transform.py`)
Invoice `vendor`	str	`r["vendor"].strip()`	None	org.data_model; tool:repo (`transform.py`)
Invoice `due_date`	date	`date.fromisoformat(r["due_date"])`	None	org.data_model; tool:repo (`transform.py`)
Invoice `amount`	Decimal, 2 dp	`Decimal(r["amount"])`	None	org.data_model; tool:repo (`transform.py`); record:memory (D-2026-044)

Element	Registered model (org.data_model)	Code today (`transform.to_model`)	Gap	Source
Invoice `paid`	bool	`r.get("paid", "false").strip().lower() == "true"`	None	org.data_model; tool:repo (`transform.py`)
Invoice `paid_date`	date or None, "added in feed v3"	**Not produced**; the module docstring lists only `id, vendor, due_date, amount, paid`	Registered, not yet implemented; implemented by TC-1 (TSK-01) in a later Data Engineering coding step	org.data_model; tool:repo (`transform.py`); task.components (TC-1, section 3)
Rejected row	Not described	`{"row": r, "reason": type(e).__name__}`	`reason` gains the value `paid_date` (section 2.3)	tool:repo (`transform.py`); task.components (TC-1)
Delay history	Not described	Does not exist	New derived structure (section 2.4)	task.components (TC-2); journey:feat.design (DC-2, B1.3)

Package version today: `0.4.0` (tool:repo, `__init__.py`).

2. Data model changes

2.1 Source contract: vendor invoice feed row (v2 → v3)

Column	v2	v3	Type in the feed	Rule	Source
`invoice_id`, `vendor`, `due_date`, `amount`, `paid`	Present	Present, unchanged	str (CSV text)	Unchanged	task.components (section 1, IN-1); tool:repo (`tests/test_pipeline.py`)
`paid_date`	Absent	**Added, optional**	ISO date string or blank	Missing key (v2) or `None` (short row) or blank → no date; ingestion needs no code change because `csv.DictReader` keys rows by header	task.components (IN-1, IN-2); journey:feat.design (B1.3 "Feed row")

2.2 Entity Invoice (DC-1): no new field beyond the registered model

Field	Type	Nullable	Derivation from the feed row	Change	Source
`id`	str	No	`invoice_id`, stripped	None	org.data_model; tool:repo
`vendor`	str	No	`vendor`, stripped	None	org.data_model; tool:repo
`due_date`	date	No	ISO date (no `strip()` added)	None	org.data_model; task.components (rules carried for coding)
`amount`	Decimal, 2 dp	No	`Decimal(...)`; never float	None	org.data_model; record:memory (D-2026-044, L-011)
`paid`	bool	No	`paid` text equals `true`, case-insensitive; missing → false	None	org.data_model; tool:repo

Field	Type	Nullable	Derivation from the feed row	Change	Source
`paid_date`	date	**Yes (None)**	`None` when missing, `None` or blank; otherwise the ISO date after `strip()`; a non-blank invalid value rejects the row	**Implemented with TC-1** (already registered)	org.data_model; journey:feat.design (B1.2 `_parse_paid_date`, B1.4); task.components (TC-1)

Rules for the record (from journey:feat.design; DD-6 and DD-7 are still marked pending there):

- The record stays a plain `dict`, not a class; every existing key keeps its name and type; `paid_date` is present on **every** accepted record, including feed v2 rows (value `None`) (journey:feat.design DC-1; task.components TC-1).
- `paid_date` is validated only after the existing field checks pass, so an existing reject reason wins over `paid_date` (journey:feat.design DD-6).
- An unpaid record may carry a valid `paid_date`; it is kept (interim, pending OQ-B6) (journey:feat.design DD-7).
- A paid record may have `paid_date = None`; it is kept and is not rejected (journey:feat.design B1.4).
- No rule requires `paid_date` to be on or after `due_date`; an earlier date gives a negative delay that is kept (journey:feat.design DC-2).

2.3 Rejected row: new reason value

Field	Type	Change	Source
`row`	dict (the feed row)	None	tool:repo (`transform.py`); journey:feat.design (B1.3)
`reason`	str	**New value** `paid_date` (constant `PAID_DATE_REASON`) beside today's exception type names (`KeyError`, `ValueError`, `InvalidOperation`, `AttributeError`)	task.components (TC-1); tool:repo (`transform.py`)

The data-quality report needs no change: `reject_reasons` is the sorted set of every rejected row's `reason`, so `paid_date` appears in it on its own (task.components IN-4; tool:repo `dq.py`).

2.4 New derived structure: Delay history (DC-2)

Attribute	Definition	Source
Producer	`delays.delays_by_vendor(records)`, a new pure module with no I/O	task.components (TC-2); journey:feat.design (B1.2)
Shape	`Dict[str, List[int]]`: key is the Invoice `vendor`, value is that vendor's list of `delay_days`	journey:feat.design (DC-2, B1.3)
`delay_days`	`(paid_date - due_date).days`, calendar days, int, may be negative	journey:feat.design (B1.2, NF-2, DC-2); record:memory (D-2026-031)
Membership	Only records where `paid` is truthy **and** `paid_date` is not `None`; a vendor with no such record has no key	journey:feat.design (B1.2, B1.5)
Order	Values in feed order; vendors in first-seen order	journey:feat.design (B1.2)
Inputs not modified	The record list is not changed	journey:feat.design (B1.2)
Persistence	None; in memory only. No storage, database or schema migration	task.components (IN-6); journey:feat.design (A1)

Attribute	Definition	Source
Consumer	ap_forecast: one vendor's list is the `history` argument of `vendor_risk`	journey:feat.design (DC-2)

2.5 Knowledge-graph relations added for this feature (ap_pipeline side)

From	Relation	To	Via	Source
Feed row (v3)	maps to	Invoice	`transform.to_model`	task.components (TC-1)
Feed row (v3)	rejected as	Rejected row (`reason = paid_date`)	`transform.to_model`	task.components (TC-1)
Invoice	aggregated by `vendor` into	Delay history	`delays.delays_by_vendor`	task.components (TC-2); journey:feat.design (DC-2)
Delay history	consumed by	ap_forecast `vendor_risk`	DC-2	journey:feat.design (A2, DC-2)
Invoice	consumed by	ap_aging `overdue_with_risk` (reads `id`, `vendor`, `due_date`, `amount`, `paid`; ignores `paid_date`)	DC-1	journey:feat.design (DC-1, B3.3)

3. Compatibility

Check	Result	Source
Feed v2 (no `paid_date` column) still loads	Yes: `paid_date = None` on every record	task.components (IN-1); journey:feat.design (NF-4)
Existing keys, types and return shapes	Unchanged; `to_model` signature and `(records, rejected)` shape kept	task.components (rules carried for coding); journey:feat.design (NF-4)
Downstream consumer of Invoice (ap_aging)	Unaffected; the extra `paid_date` key is ignored	journey:feat.design (B3.3)
Monetary type	`amount` stays Decimal	record:memory (D-2026-044, L-011)

4. Schema validation rules (for the later unit-test step)

Rule	Expected	Source
Every accepted record has exactly the keys `id`, `vendor`, `due_date`, `amount`, `paid`, `paid_date`	Pass	org.data_model; journey:feat.design (DC-1)
`paid_date` is a `date` or `None`	Pass	org.data_model
Blank or missing `paid_date`	`None`, row accepted	journey:feat.design (B1.4)
Non-blank invalid `paid_date` (for example `"2026-13-01"`, `"not-a-date"`) on an otherwise valid row	Rejected, reason `paid_date`, listed in `reject_reasons`	journey:feat.design (B1.4, B1.5)
Bad `due_date` and bad `paid_date` together	Reason `ValueError`	journey:feat.design (B1.5, DD-6)
Delay history values are `int` and keys are vendors with at least one paid record carrying a `paid_date`	Pass	journey:feat.design (DC-2, B1.5)
`amount` is `Decimal`, never `float`	Pass	record:memory (D-2026-044)

5. Code in this step

This step changes the data model only and changes no code. The tool view shows `transform.py` without `paid_date` and no `delays.py` module (tool:repo). The model change is implemented by TC-1 and TC-2 in the later Data Engineering coding step, within the file budgets DE-04 listed (task.components section 3). Agent judgement: the `transform.py` module docstring lists the model fields, so it should gain `paid_date` with TC-1 (tool:repo `transform.py`).

6. Open questions (non-blocking)

ID	Question	Interim answer	Owner (proposed)
DMQ-A	Should the registered org.data_model be extended with the Rejected row `reason` value `paid_date` and the Delay history structure (sections 2.3 and 2.4)? Today it describes only the Invoice entity	Treat this document as the feature-level model change; data modelers decide whether to register them org-wide	Data engineering lead; data modelers
DMQ-B	Carried CQ-I1: is the feed v3 column spelled exactly `paid_date`?	Assume `paid_date` as in org.data_model	Data engineering lead / feed owner
DMQ-C	Carried OQ-B6 (DD-7): should an unpaid record keep a valid `paid_date`?	Kept; ignored by the Delay history	Product / Architect
DMQ-D	Carried DQ-4: should `delays_by_vendor` be re-exported from `ap_pipeline/__init__.py`?	No; imported as `ap_pipeline.delays.delays_by_vendor`	Architect

7. Validation check

Check	Result	Source
Data model updated from the source system and ingested data	Pass: feed v3 contract (2.1), Invoice (2.2), Rejected row (2.3), Delay history (2.4), relations (2.5)	task.components; org.data_model
Conflict with the registered model or an approved output	None: `paid_date` matches org.data_model and DC-1; additions are new structures, not changes	org.data_model; journey:feat.design; task.components (section 5)
Production or source-system data read	None; only inputs, the tool view and synthetic test data	record:brief
Blocking questions	None	—